Algorithms and Mathematical Concepts for Computing

11th November 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

Consider the following scenario:

You are tasked with developing a software application for a library to manage their book lending system. The application needs to track book availability, update lending statuses and generate reports on lending trends.

a) State in which stage of the software development process user interface layouts are created, based on the initial requirements and defined project scope. **1**

b) State which stage of the software development process involves locating bugs and ensuring it meets the specified requirements before deployment. **1**

c) Explain the purpose and benefits of encapsulation and modularity in software design. Give ONE (1) real-life example for each encapsulation and modularity. **8**

Total 10 Marks

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Marks

Question 2

Consider the following content related to numbers and number systems:

a) Which of the following is **NOT** a type of number? **1**

i) Integer
ii) Fraction
iii) Decimal
iv) Symbol

b) Fill in the missing part of the following statement: **1**
Decimal to _____ conversion involves converting a number from base 10 to base 16.

c) Describe the concept of greatest common divisor (GCD) **and** explain its significance in number theory. **2**

d) Perform the following conversions and show your working. **2**

i) Convert the octal number 76 to binary number.

ii) Convert the hexadecimal number 2F to decimal number. **2**

iii) Convert the decimal number 100 to binary number. **2**

Total 10 Marks

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Marks

Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following operators represents the logical NOT operation? **1**

i) $\wedge$
ii) $\vee$
iii) !
iv) $\rightarrow$

b) Complete the truth table for the Boolean expression: **6**

$$(P \wedge \neg Q) \vee (Q)$$

P	**Q**	$\neg Q$	$P \wedge \neg Q$	$(P \wedge \neg Q) \vee (Q)$
T	T			
T	F			
F	T			
F	F			

c) Explain the concept of biconditional statements in Boolean algebra **and** provide ONE (1) example. **3**

Total 10 Marks

Question 4

Answer the following questions related to Predicate Logic, Set Theory and Venn Diagrams:

a) Which of the following symbols represents the subset relationship in set theory? **1**

i) $\subseteq$
ii) $\subset$
iii) $\cup$
iv) $\cap$

b) Explain the concept of set complement **and** provide ONE (1) example. **2**

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Marks

c) Calculate the union of the following sets: **1**

A = {2, 4, 6} and B = {1, 2, 3}.

d) What is the Cartesian product of two sets and how is it represented? **4**

Calculate the Cartesian product of the following sets: A = {a, b} and B = {1, 2}.

e) Calculate the intersection of the following sets and represent your answer using the intersection symbol (∩): **2**

A = {5, 10, 15} and B = {10, 20, 30}.

Total 10 Marks

Question 5

Answer the following questions related to statistics:

a) Discuss the concept of standard deviation as a measure of variability. Provide ONE (1) example that is appropriate and ONE (1) example that is inappropriate for using standard deviation. **3**

b) Fill in the missing part of the following statement: **1**
In statistics, _____ is a measure that quantifies the variability of a data set by identifying the difference between the smallest and largest values.

c) Calculate the range of the following data set: 12, 7, 15, 10, 5. Show your working. **2**

d) Explain the concept of skewness and why it is useful. **2**

e) Calculate the mode of the following data set: 3, 6, 3, 8, 9, 3, 10. Show your working. **2**

Total 10 Marks

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Marks

Question 6

Answer the following questions related to Data Structures:

a) Which one of the following is **NOT** a characteristic of a queue? **1**

i) First-In-First-Out (FIFO) structure.
ii) Allows random access to elements.
iii) Useful for scheduling tasks.
iv) Can dynamically grow or shrink in size.

b) State the name of a data structure in which elements can be accessed in a random manner. **1**

c) Define the data type "stack" **and** explain its role in managing data. **2**

d) Write pseudocode for a function to insert an ***element*** into a ***queue***. The ***element*** and the ***queue*** should be parameters of the function. **4**

Assume the existence of a function called ***size(queue)*** that returns the number of elements in the queue, and assume the existence of a function called ***append(queue, element)*** that adds an element to the queue.

Check that the size of the queue does not exceed MAX_SIZE (a predefined constant). If it exceeds MAX_SIZE, print an error message.

Total 10 Marks

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Marks

Question 7

Answer the following questions related to Data Structures:

a) Which data structure is typically used for implementing function calls, allowing for the storage of local variables and control flow? **1**

i) Array
ii) Queue
iii) Stack
iv) Hash Table

b) Fill in the missing parts of the following statement: **2**
In a queue, elements are added to the _____ and removed from the _____.

c) Consider a stack initially containing the elements 100, 200, 300. Assume 100 is at the top of the stack. Perform the following operations one after the other: **3**

i) Pop one element.
ii) Push the number 300.
iii) Pop two elements

Show the final contents of the stack after these operations.

Show your working and draw out the stack after each operation. Clearly identify the top of the stack.

d) Provide ONE (1) real-world example and justification of how each queue data structure and stack data structure would be used. **4**

Total 10 Marks

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Marks

Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following best describes time complexity in algorithm analysis? **1**

i) The minimum amount of time an algorithm requires to complete.
ii) The measure of the algorithm's runtime as the input size grows.
iii) The average-case scenario of an algorithm's efficiency.
iv) The number of operations performed by an algorithm for a given input size.

b) Fill in the missing part of the following statement: **1**
The best case time complexity of binary search algorithms is _____.

c) Write pseudocode for a function that implements a binary search algorithm to find a target value in a sorted array. The function takes the target value and array as parameters. It returns the index of the target value or -1 if the target is not found. **6**

You should assume the existence of a function called ***length(arr)*** which returns the number of elements in the array ***arr***.

d) Explain why a hash table can be more efficient than a binary search tree for lookups. **2**

Total 10 Marks

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Marks

Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which sorting algorithm repeatedly compares adjacent elements and swaps them if they are in the wrong order? **1**

i) Merge Sort
ii) Bubble Sort
iii) Insertion Sort
iv) Quick Sort

b) State what is meant by backtracking in programming. **2**

c) Fill in the blanks in the pseudocode for the factorial function implementing recursion: **5**

```
Function factorial(n):
   if _____ == 0:
      return _____
   return _____ * _____  (_____)
End function
```

d) Explain how recursion can be applied to traversing tree structures. **2**

Total 10 Marks

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Marks

Question 10

Answer the following questions related to tree data structures:

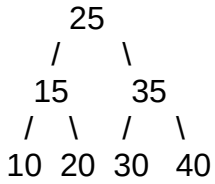
a) Which of the following best describes a binary tree? **1**

i) A tree structure where each node has at most two children.
ii) A tree structure where every node has exactly two children.
iii) A tree structure that stores elements in a linear fashion.
iv) A hierarchical structure where nodes are connected in a circular manner.

b) Fill in the missing part of the following statement: **1**
In a binary tree, a leaf node is defined as a node with _____ children.

c) Construct a binary search tree from the following sequence of numbers: 40, 20, 60, 15, 30, 50, 70. Ensure that 40 is the root of the tree. **4**

d) Using the provided binary search tree (BST) below, demonstrate the steps involved in searching for the value 30. Begin the search at the root node and describe each comparison and movement until the value is found or determined to be absent. **4**



Total 10 Marks

End of paper

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